COURSE

IB Diploma Programme Business management HL

Production and capacity

Unit 5: Operations management · 5.2 Operations methods; 5.6 Production planning (HL only)

Revision for Unit 5: Operations management: each firm works it out on its own deals.

Class	Topic
IB Diploma Programme Business management HL	**Production and capacity**

THE LESSON

Setup	2 min

Round 1 — Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	**Deals**	TASK 1 **Capacity used %**	TASK 2 **Full capacity risks**

Round 2 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 3 **Fixed cost per car**	TASK 4 **New productivity**	**B2B deal**

Round 3 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 5 **Meet a short peak**	TASK 6 **Unit cost when full**	**B2B deal**

Final questions	3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

Workshop capacity.

Capacity used, as a %?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Prep bays	**3 bays**
Most one bay can do a week	**10 cars**
Cars prepared a week	**24 cars**

METHOD

Capacity used = cars prepared ÷ (bays × most per bay) × 100

3 × 10 = 30; 24 ÷ 30 × 100 = 80%

Why: It has 6 cars of spare capacity a week.

Task 2

ROUND 1

You run at 100% capacity all year.

Which two problems are most likely?

PICK TWO

- ○ A higher fixed cost per car
- ✔ **No spare time for repairs**
- ○ Spare capacity is wasted
- ✔ **Staff stress and more errors**
- ○ Staff with too little to do

Why: Full capacity spreads fixed costs well, but leaves no slack.

Task 3

ROUND 2

Three workshops: fixed costs a year, and cars prepared.

Order by fixed cost per car, lowest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£45,000 fixed, 500 cars**
2. **£30,000 fixed, 300 cars**
3. **£20,000 fixed, 160 cars**

Why: £90, £100, £125 a car: the biggest costs, spread widest, are cheapest.

Task 4

ROUND 2

More staff, more cars.

New cars prepared per member of staff?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cars prepared a week	**18 cars**
Staff	**3 staff**
Extra staff	**3 staff**
Extra cars prepared a week	**30 cars**

METHOD

Per staff = (cars + extra cars) ÷ (staff + extra staff)

(18 + 30) ÷ (3 + 3) = 8 cars

Why: Productivity rose from 6 to 8 cars per worker.

Task 5

ROUND 3

Demand is above your capacity for two months only.

How should you meet it, and why?

PICK A OR B AND GIVE A REASON

- ○ Build a new workshop
- ✔ **Subcontract the extra work**

Why: A short peak does not pay for new capacity.

Task 6

ROUND 3

Run the workshop full.

Fall in unit cost at full capacity?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Fixed costs, a month	**£12,810**
Variable cost per car	**£400**
Cars prepared a month	**15 cars**
Most it can prepare a month	**35 cars**

METHOD

Fall = fixed costs ÷ cars now − fixed costs ÷ most cars

£12,810 ÷ 15 = £854; ÷ 35 = £366; £854 − £366 = £488

Why: The variable cost stays; only the fixed cost per car falls.

Final questions

6
questions for each pupil

109
Level 3 Business questions

10
of them on production and capacity

The game draws 6 questions on this topic.

1 A service centre can complete 640 jobs a month but completes 512. What is its capacity utilisation?
80%.

2 A parts team produces 1,440 tested units in 240 labour hours. What is labour productivity?
6 units per labour hour.

3 A dealer considers outsourcing parts deliveries. Which comparison best supports an operational decision?
Total cost, reliability, lead-time risk and capability kept or lost.

4 A bodyshop's practical monthly capacity is 900 repair-hours and booked work uses 765 hours. What is spare capacity as a percentage of practical capacity?
15%.

5 A workshop raises utilisation from 82% to 98%, while waiting time and rework rise sharply. What is the best interpretation?
It is near its limit, so delays rise.

6 Technician productivity differs widely between branches. Which study is most likely to support a fair operational decision?
Compare job mix and hours first.

7 Before a wellbeing redesign a workshop produced 450 standard hours with 90 paid hours; after, 520 with 104. What happened to measured labour productivity?
It remained 5 standard hours per paid hour.

Final questions

8 A fitting centre's 27 fitters fit 918 tow bars a month. What is labour productivity?

34 tow bars per fitter.

9 A bodyshop can do 250 repairs a month and did 160. What is its capacity utilisation, and the effect on unit costs?

64%: each repair carries more fixed cost than at full capacity.

10 A workshop can service 400 vehicles a month and serviced 300 last month: 75%. Which measure is that?

Its capacity utilisation.